Shaikh Muhammad Zain

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Summary

AI-specialized Computer Science student at NED University (graduating 2027), seeking ML engineer internships. Builds deep learning, vision and LLM systems, tests them on held-out data with ablations, and deploys them.

Skills

- **Machine Learning:** supervised learning, clustering (K-Means, Gaussian mixtures), cross-validation, hyperparameter tuning (Optuna), SHAP, fairness auditing, bootstrap significance testing
- **Deep Learning and Vision:** PyTorch, Vision Transformers, self-attention, face detection and alignment, ArcFace embeddings, fine-tuning, mixed-precision training
- **LLMs and Retrieval:** RAG, LangGraph, BM25, Hugging Face, Gemini API, MCP, prompt-injection checks
- **Languages and Libraries:** Python, scikit-learn, XGBoost, Pandas, NumPy, PostgreSQL
- **Deployment and MLOps:** FastAPI, Flask, Docker, MLflow, GitHub Actions, pytest, Vercel, Render

Experience

Machine Learning Intern, *FlyRank* — Jul 2026 – Aug 2026 | *Remote*

- Clustered 18,752 production web pages from 29 client sites into content archetypes with K-Means, compared against Gaussian mixture and agglomerative baselines, and turned them into an action playbook per archetype.
- Validated stability with leave-one-client-out testing (mean ARI 0.909, 17 clients) and a feature-ablation test; withdrew the one archetype that failed and documented the limits in a published paper.

Artificial Intelligence Intern, *DecodeLabs* — Jun 2026 – Jul 2026 | *Remote*

- Built two Python apps with Flask APIs, tests in CI and live demos: a KNN classifier (96.0% 5-fold CV accuracy) that shows the neighbors behind each prediction, and a chatbot with five-tier intent matching, including TF-IDF.

Information Technology Intern, *Karachi Development Authority (KDA)* — Feb 2026 – Mar 2026 | *Karachi*

- Resolved day-to-day hardware and software issues for the IT department, working with the wider team in a government office environment.

Projects

Face Recognition: Runner-up of 50 university teams, NED CSIT TechFest '26 — *PyTorch, InsightFace, ONNX*

- Built the team's offline CPU face recognition pipeline (detection, alignment, quality filter, ArcFace embeddings, tuned threshold): 97.1% accuracy, 2.9% EER on 40 held-out South Asian people; without alignment, 25.4% EER.
- Fine-tuned ArcFace on a cloud GPU with person-disjoint splits and a bootstrap over people; it did not clearly beat the original, so the team kept the original model and reported the result.

Leash: Payment Guardrails for AI Agents — *Gemini, PayPal SDK | GitHub | Demo*

- Built the full service for a team entry in the PayPal AI Hackathon 2026: AI agents (Gemini, Claude via MCP) pay via PayPal only within user-set rules, enforced by code checks on budget, store, price and duplicates.
- Added a Gemini review that detects prompt injection on shop pages; it can block or escalate a payment but never approve one the code rejected, and any AI or PayPal failure blocks the payment.

ViT Paper Reproduction: Mimetic Initialization (ICML 2023) — *PyTorch, CIFAR-10 | GitHub*

- Reproduced the paper in a team of three with no official code: wrote a 5.3M-parameter Vision Transformer, the SVD-based mimetic initialization and the training loop from scratch in PyTorch, with mixed-precision training.
- Confirmed the paper's claim: mimetic initialization raised CIFAR-10 test accuracy from 85.13% to 88.87% (+3.74 points, 79% of the reported gain), and it beat the 100-epoch baseline after only 50 epochs.

FYP: StepGuard, Hallucination Detection for Agentic RAG — *LangGraph, Transformers, HotpotQA*

- Co-building a system that checks each step of a multi-hop RAG agent against its evidence, finds the first failing step and triggers an automatic repair; proposal approved by the FYP panel.
- Co-built the LangGraph agent testbed on HotpotQA and an error-injection pipeline for labeled traces; measured the baseline to beat on a 67-trace pilot (LettuceDetect: step F1 0.800, 1.17 s per step).

EduPulse: Student Performance Intelligence — *FastAPI, MLflow | GitHub | Demo*

- Benchmarked ten model families with Optuna tuning for at-risk prediction and score regression (hold-out R^2 0.882), with SHAP, a fairness audit and drift monitoring, served via FastAPI and Docker with 63 tests in CI.

Education

NED University of Engineering and Technology, *Karachi* — Oct 2023 – Aug 2027

BS Computer Science and Information Technology (Specialization: Artificial Intelligence) | **CGPA: 3.4/4.0**

DJ Sindh Government Science College, *Intermediate (Pre-Engineering)* — 2022

Certifications

Google AI Professional Certificate, Coursera (Jun 2026) | FlyRank ML and AI Fluency Certifications (2026)